

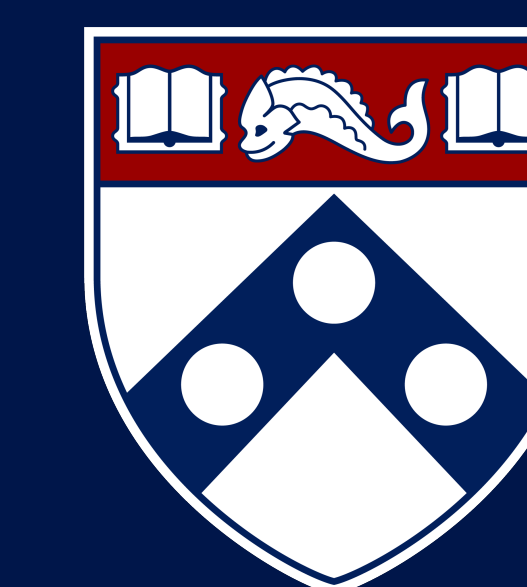


The Language
Learning Lab

Observing negation: Naturalistic and artificial Human Simulation Paradigms

Victor Gomes¹, Yubin Huh¹, Roman Feiman², Susan Goldin-Meadow³, John Trueswell¹

¹University of Pennsylvania, ²Brown University, ³University of Chicago



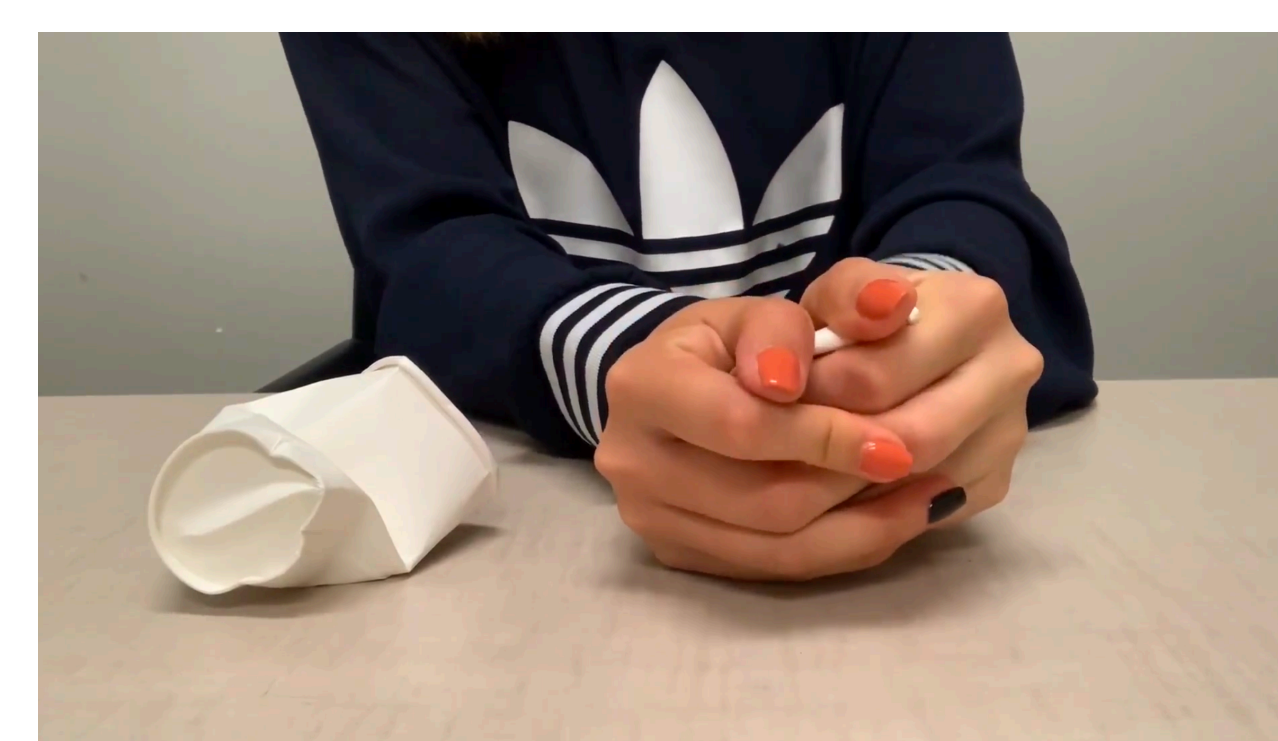
mindCORE
ILST

Background

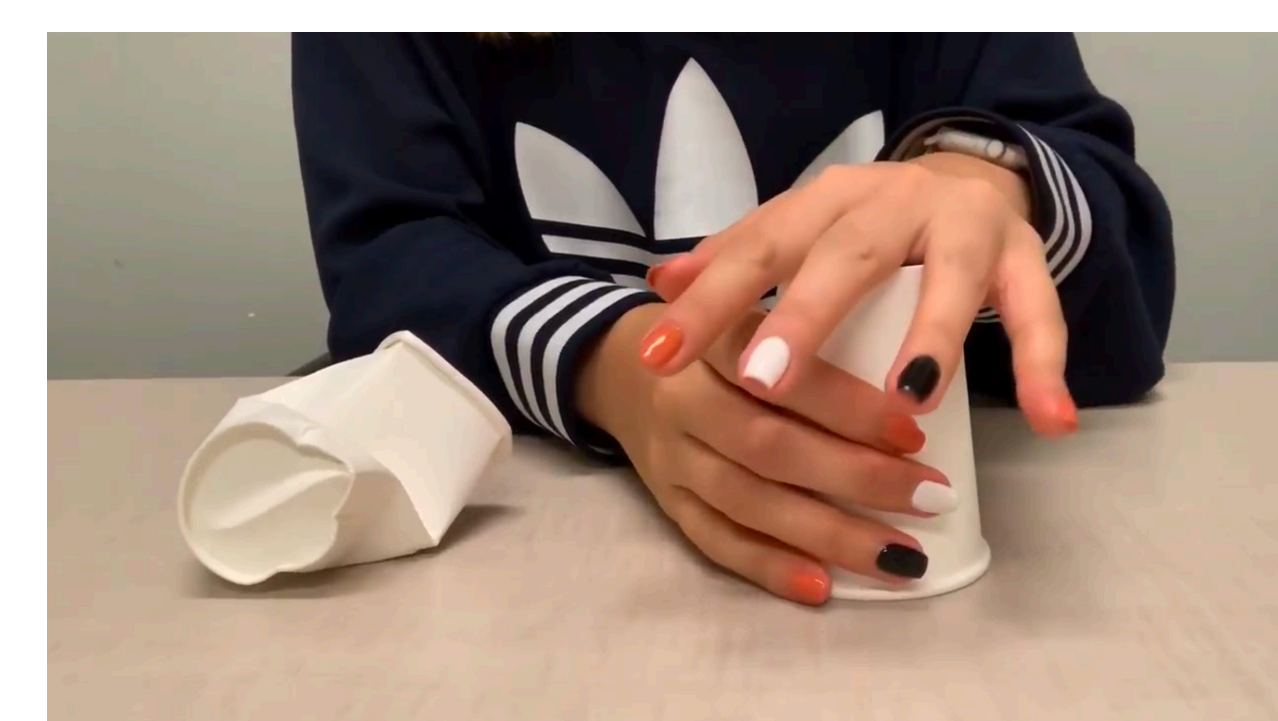
- Children have difficulty with negation before 2 years
- Struggle with “no” & “not” without pragmatic support (Nordmeyer & Frank, 2014; Austin et al., 2014; Feiman et al., 2017; Reuter, Feiman, & Snedeker, 2018; de Carvalho, Crimon, Barrault, Trueswell & Christophe, in revision)
- Produce utterances with “not” recognizable as “truth-functional” around 30 months (Bloom 1968; Choi, 1988)
- Does delay reflect conceptual or linguistic limitations?
- Q1: Are there situations that make negators easier to infer in a message?**
- If delay is due to abstractness, then specific contexts may make negation more available as inference.
- Q2: To what degree can linguistic cues assist in such contexts?**
- Converging linguistic cues should boost performance
 - But, since there is a delay, syntax or lexical information alone may not be enough.

Methods

Watch videos and guess how parents would describe them



SUCCESS

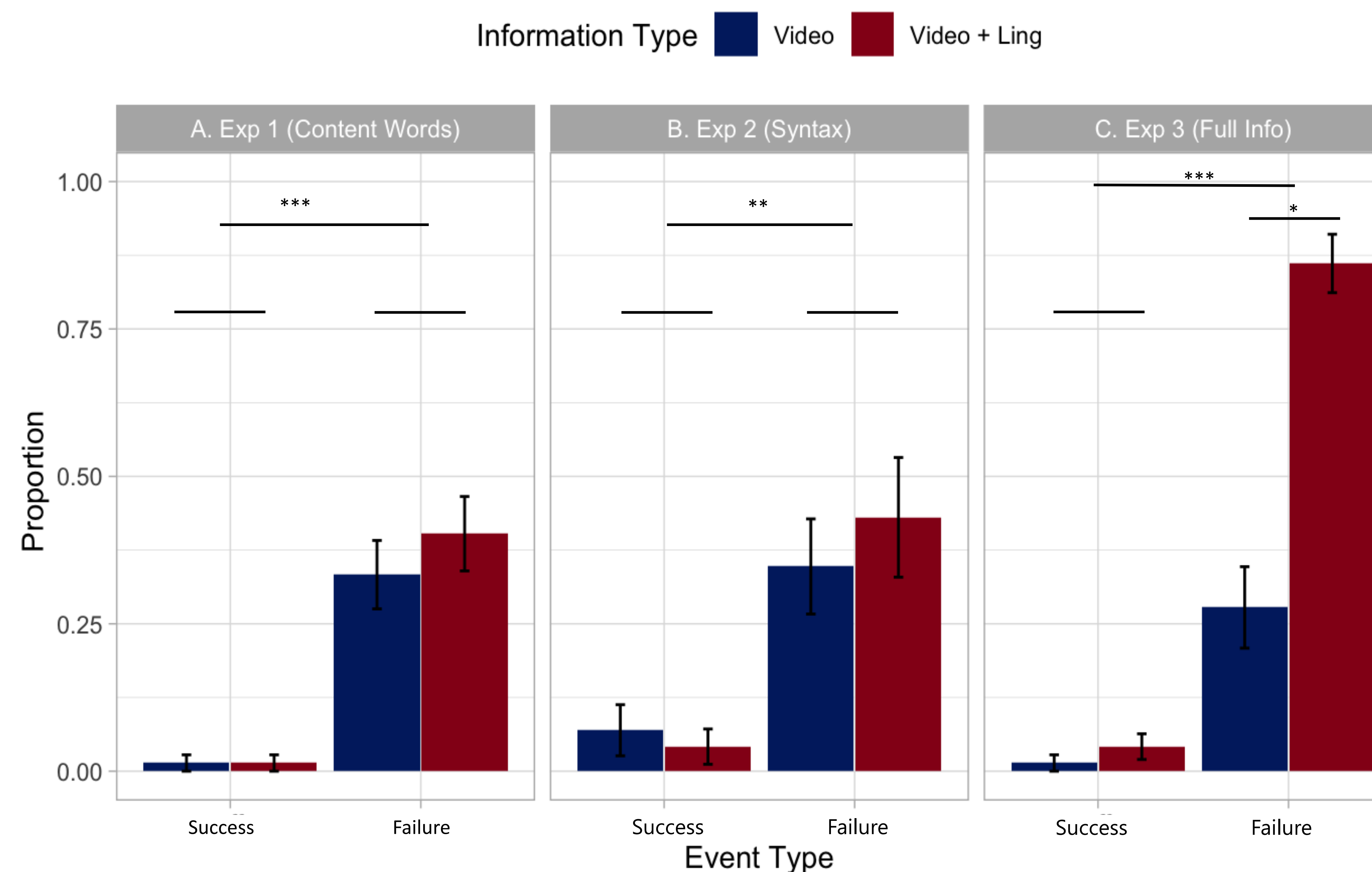


FAILURE

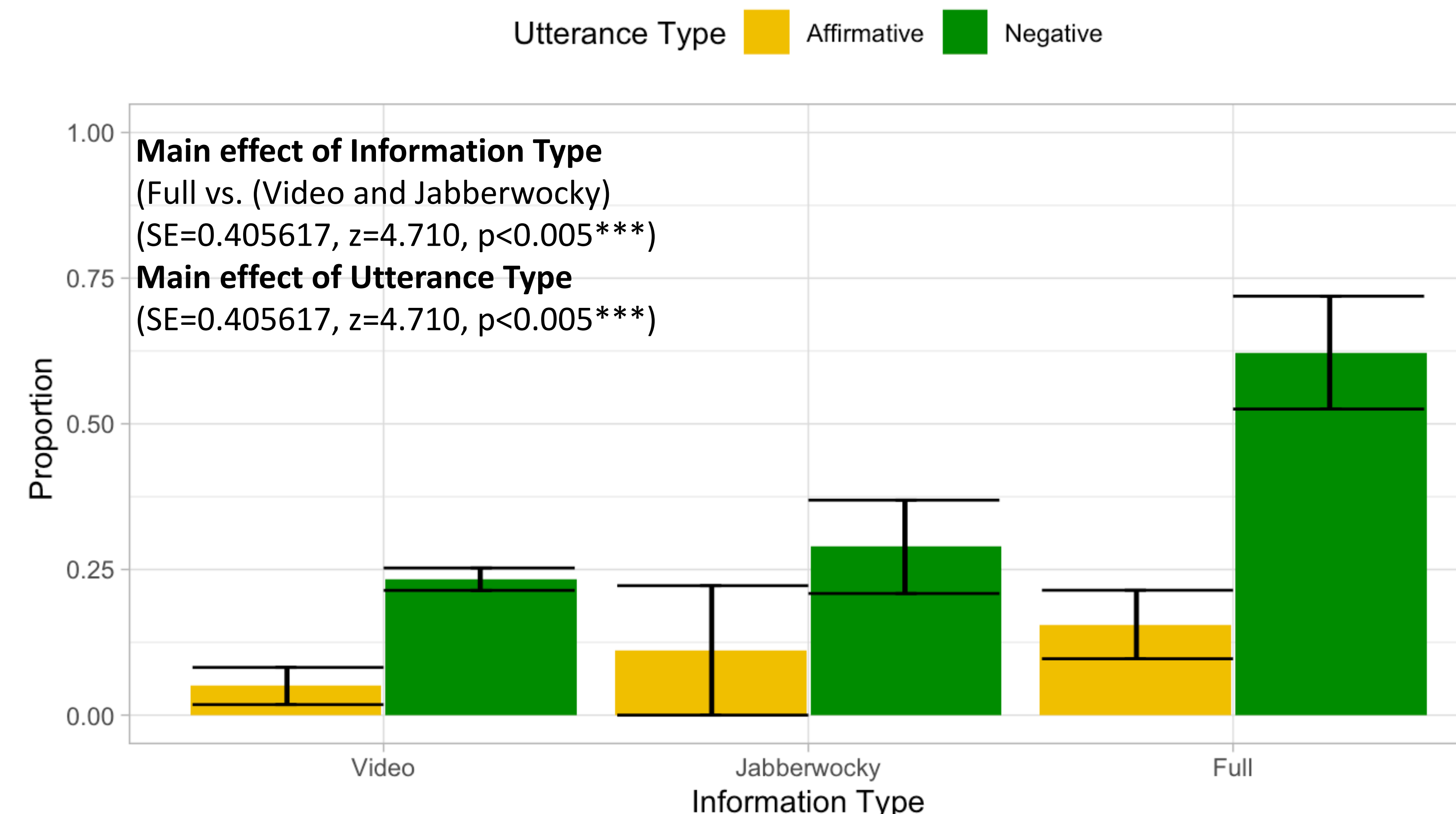
Between participant condition:
video alone, or video with
language cue after video ends

- Linguistic cue varied by experiment
 - Exp. 1: Syntax only (w/ Jabberwocky)
 - Exp. 2: Content Words only (alphabetized)
 - Exp. 3: Full Language context (“fill in the blank”)
- n=36 adult mturkers per Exp., hosted on pCIBEX

Results



Naturalistic Pilot Results



Naturalistic Pilot

n=10 for in-lab pilot

- 20 vignettes randomly sampled from 4 families in LDP Corpus (children were 14-22mo)
- All videos muted, with a continuous tone inserted over target utterance
- 10 target utterances had negators, 10 did not
- 3 Between-Subject conditions
 - Video only
 - Video + Jabberwocky
 - Video + Full Language context

General Discussion

- Q1: Are there situations that make negators easier to infer in a message?**
 - Yes, scenes of failure can encourage the inference of negation in an utterance.
 - Robust across all experiments
- Q2: To what degree can linguistic cues interact with such contexts?**
 - For partial cues (Exp. 1 & Exp. 2), no effect above context
 - For full cues (Exp. 3), there is a reliable effect and interaction with context
- Specific contexts in which negators are uttered can significantly and reliably assist inference of negator in conceptually-equipped learners
 - Pilot indicates this is true of naturalistic scenes as well
- Linguistic cues cannot support inference of negation alone, but can when integrated with context and other linguistic cue.